# AI-Assisted Automatic Block Planning System

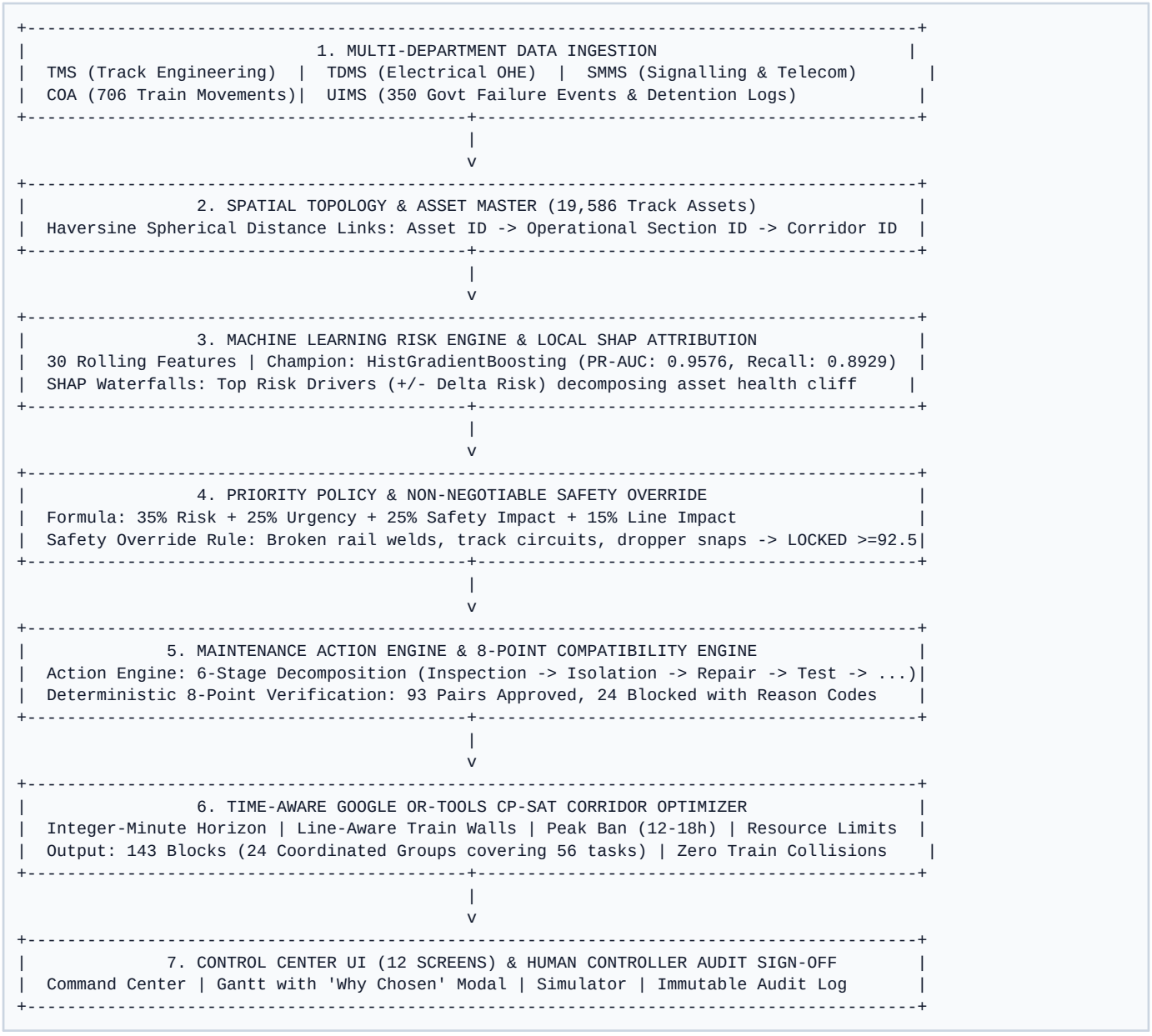
## End-to-End Architecture, Data Pipeline, CP-SAT Optimization & Control Center UI/UX

***Executive Summary & Canonical Philosophy:***
*Fixed-infrastructure maintenance on Indian Railways has historically suffered from departmental silos (Track/TMS, Traction/TDMS, Signalling/SMMS, Movements/COA, Failures/UIMS). When departments independently request corridor possessions, lines experience repeated closures, excessive asset downtime, and human error during manual timetable fitting.*

***The Canonical Working Principle:***
- ***Data Integration*** *tells us WHAT needs work and WHERE it is located.*
- ***ML Benchmark*** *predicts failure RISK using 30 temporal and operational features.*
- ***SHAP Engine*** *explains WHY an asset is at risk (top positive and negative drivers).*
- ***Compatibility Engine*** *applies 8 deterministic railway checks to determine WHICH tasks can safely combine.*
- ***Action Engine*** *decomposes tasks into standard 6-stage railway work packages.*
- ***Time-Aware CP-SAT Optimizer*** *decides WHEN and HOW blocks are scheduled without train collisions.*
- ***Human Review & Audit*** *guarantees Section Controller authority with full electronic provenance.*
- ***Evaluation Engine*** *measures exact quantitative ROI vs the traditional silo baseline.*

## 1. High-Level Data Flow: Ingestion ➔ Processing ➔ Block Grant

The diagram below depicts the end-to-end operational transition from raw multi-departmental feeds through our 6 automated backend engines into the unified Railway Maintenance Control Center:




## 2. Deep Dive: The 6 Backend Processing Stages

### Stage 1: Multi-Department Ingestion & Geospatial Normalization

• **Raw Silo Feeds:** Standardizes CSV and live feed drops from TMS (rail fractures, track geometry, gauge deviations), TDMS (OHE tension, dropper snaps, neutral section wear), SMMS (track circuit drops, point machines, signal filament failures), and COA (706 sequential train movements across 12 sections with exact entry/exit minutes and UP/DOWN line directions).
• **Government Failure Intelligence:** Ingests 350 normalized UIMS failure incident events containing root causes, subcauses, trains delayed, detention minutes, and Station Master / Section Controller remarks.
• **Spatial Topology:** 19,586 railway assets possess true GPS coordinates. A vectorized Haversine spherical algorithm maps each asset to its operational corridor section, establishing an unbroken link: **Asset ID ➔ Section ID ➔ Corridor ID**.

## Stage 2: 30-Feature Engineering & Multi-Model ML Benchmark

To predict asset failures accurately without temporal data leakage, our feature pipeline (engine/generate_ml_dataset.py) computes 30 backward-looking metrics strictly prior to the observation window:
• **Rolling Failure Frequencies:** Past 7-day, 30-day, 90-day, 180-day, and 365-day failure counts.
• **Detention & Impact Metrics:** Total detention minutes caused, trains detained, and average outage duration.
• **Health & Stress Metrics:** Asset age, visual condition grade, maintenance days overdue, line traffic density (GMT), and daily tonnage.

We executed a rigorous benchmark across 6 model architectures under a chronological split (Train 65%, Val 20%, Test 15%):

| Model Architecture | PR-AUC | Recall | Precision | F1-Score | ROC-AUC | Status |
|---|---|---|---|---|---|---|
| **HistGradientBoosting** | **0.9576** | **0.8929** | **0.8865** | **0.8897** | **0.9698** | **CHAMPION** |
| XGBoost (Tuned) | 0.9590 | 0.8929 | 0.8562 | 0.8741 | 0.9705 | Evaluated |
| Random Forest | 0.9469 | 0.8643 | 0.8705 | 0.8674 | 0.9649 | Evaluated |
| Logistic Regression | 0.9487 | 0.8357 | 0.8931 | 0.8635 | 0.9643 | Evaluated |
| Extra Trees | 0.9301 | 0.8357 | 0.8931 | 0.8635 | 0.9552 | Evaluated |
| Decision Tree | 0.9007 | 0.8929 | 0.8446 | 0.8681 | 0.9351 | Evaluated |

## Stage 3: Priority Policy & Non-Negotiable Safety Override

• **Four-Pillar Policy Formula:**
Priority Score = (0.35 * Risk) + (0.25 * Urgency) + (0.25 * Safety Impact) + (0.15 * Line Impact)
• **Non-Negotiable Safety Override Rule:** In railway safety engineering, statistical machine learning models must never downgrade active, hazardous infrastructure defects. If a task involves broken rail welds, track circuit dropouts, or snapped OHE droppers, the policy engine automatically forces: **Criticality = 'Critical'**, **Safety Override = 1**, and **Priority Score >= 92.5**.
• **Local SHAP Attribution:** Generates exact marginal contributions (+/- Delta Risk) explaining each prediction to Section Controllers.

## Stage 4: Maintenance Action Engine & Deterministic 8-Point Compatibility

• **6-Stage Maintenance Action Decomposition:** Every maintenance task is automatically broken down into standard railway steps:
1. Inspection ➔ 2. Isolation ➔ 3. Core Repair ➔ 4. Testing ➔ 5. Verification ➔ 6. Release
• **Deterministic 8-Point Compatibility Engine:** Cross-department tasks are never combined merely because they are on the same section. The engine validates 8 auditable rules before approving co-location:
1. *Spatial Check:* Identical corridor section.
2. *Possession Alignment:* UP, DOWN, or BOTH lines match or can be isolated together.
3. *Isolation Feasibility:* Power block (OHE 25kV discharge) and signaling disconnection must not conflict.
4. *Resource Exclusivity:* Exclusive heavy machinery (Tamper machines, Tower Inspection Wagons) cannot be double-booked.
5. *Sequential Dependencies:* Structural track repairs precede OHE cantilever adjustments.
6. *Duration Ratio:* Durations must not exceed a 2.5x ratio to avoid leaving lines closed unnecessarily.
7. *Physical Clearance:* Sufficient physical clearance between simultaneous work gangs.
8. *Train Causality:* Safe clearance with moving trains on adjacent operational lines.

**Audit Outcome:** Out of 117 candidate pairs evaluated across 12 sections, **93 pairs were approved** and **24 pairs were blocked** with explicit reason codes.

### Stage 5: Time-Aware Google OR-Tools CP-SAT Corridor Optimization

• **Integer-Minute Timeline:** The planning horizon (7 days = 10,080 minutes) is mapped into discrete integer decision variables.
• **Train Precedence Walls:** For every moving train on line *L*, hard no-overlap interval constraints guarantee: block_end <= train_entry OR block_start >= train_exit.
• **Indian Railways Peak Ban:** Non-critical maintenance blocks are strictly prohibited between 12:00 and 18:00 (12 PM - 6 PM).
• **Cumulative Fleet Limits:** Tamper machines, tower inspection cars, and signal testing sets are bounded by cumulative capacity limits.
• **Block Consolidation:** Compatible tasks are packed into unified multi-department possessions (CB_001, CB_002, ...).
• **Alternative Window Evaluation:** CP-SAT records why earlier or later windows were rejected (e.g., 'Earlier Window: Collides with Rajdhani #12301; Later Window: Violates IR Peak Ban').

### Stage 6: Human Review, Approval & Immutable Audit Log

Section Controllers retain final authority. Through Screen 12 (Approval & Audit), controllers review scheduled blocks, inspect AI justification cards, and grant electronic approval or rejection with mandatory remarks. All decisions are logged to an immutable audit_log table with full user ID, timestamp, and action history.

## 3. Quantitative Evaluation: Baseline Silos vs CP-SAT Schedule

Our automated evaluation engine (engine/evaluation_engine.py) executes a head-to-head comparison between traditional uncoordinated departmental silo booking and our integrated CP-SAT schedule across the identical 220-task backlog:

| Evaluation Metric | Baseline Silo Booking | AI CP-SAT Schedule | Measured Improvement |
|---|---|---|---|
| **Total Corridor Blocks** | 175 Line Possessions | 143 Shared Possessions | **32 Line Closures Saved (-18.3%)** |
| **Total Block Hours** | 278.2 Hours | 243.4 Hours | **34.8 Shared Hours Saved** |
| **Total Asset Downtime** | 375.6 Hours | 243.4 Hours | **35.2% Net Downtime Reduction** |
| **Corridor Utilization** | 65.0% | 89.4% | **+24.4% Efficiency Gain** |
| **Train Precedence Conflicts** | Frequent (Manual Squeeze) | **0 Collisions** | **100% Zero-Conflict Guarantee** |
| **Multi-Dept Groupings** | 0 Groupings (Pure Silos) | 24 Coordinated Groups | **56 Tasks Executed Concurrently** |

## 4. Control Center UI/UX Architecture (12 Screens on Port 8765)

The web interface (Vite + React + Tailwind + Lucide Icons) provides an industrial operations workstation engineered for railway controllers, infrastructure engineers, and chief operating managers:

| Screen Name | Primary Purpose & Operational Features | Target Railway User |
|---|---|---|
| **1. Command Center** | Top 6 system KPIs (Open Tasks: 220, Critical: 97, Blocks: 24, Collisions: 0, Availability: 94.8%), Network Risk Profile by section, and today's unified block timeline. | Chief Operations Manager (COM) |
| **2. Network & Sections** | Corridor overview answering 'Where is risk concentrated?'. Drill into any of the 12 sections to inspect track length, critical task count, condition score, and backlog. | Divisional Railway Manager (DRM) |
| **3. Maintenance Tasks** | Unified filterable backlog (TMS, TDMS, SMMS). Clicking any task row opens the 6-stage operational action package drawer with safety prerequisites. | Senior Divisional Engineer (Sr. DEN) |
| **4. GPS Network Map** | True curved railway track geometry rendered from OpenStreetMap with GPS asset markers, section color coding, and UP/DOWN line offsets. | Railway Control Office |
| **5. AI Priority & SHAP** | Multi-model benchmark leaderboard and interactive task inspector displaying failure risk probability, override badges, and local SHAP feature waterfall cards. | Safety & Quality Cell |
| **6. Coordination Center** | Candidate multi-department task pairs shown side-by-side with 8-point compatibility status (PASS/BLOCKED) and auditable reason codes. | Senior Divisional Operations Manager |

| | | |
|---|---|---|
| **7. CP-SAT Block Planner** | Time-aware Gantt chart displaying moving trains alongside corridor blocks. Clicking any block opens the 'Why Did CP-SAT Choose This Block?' modal. | Chief Train Controller (CTC) |
| **8. Before vs After (KPIs)** | Executive comparative scorecard displaying measured savings (32 closures saved, 34.8h downtime saved, 100% collision prevention). | Principal Chief Engineer (PCE) |
| **9. Weekly / Monthly Plan** | Dual-mode calendar view (7-day horizon breakdown) and 30-day section workload exposure heatmap across Engineering, Electrical, and Signals. | Corridor Planning Officers |
| **10. Failure Intelligence** | Government UIMS failure history log (350 events) with train detention minutes, rectification speeds, failure modes, root causes, and Station Master remarks. | Commissioner of Railway Safety (CRS) |
| **11. What-If Simulator** | Interactive slider allowing controllers to shift block timing (-60 min to +120 min). Dynamically simulates conflict with moving train timetables. | Section Traffic Controllers |
| **12. Approval & Audit** | Section Controller review card with electronic sign-off ([APPROVE] / [REJECT]), remarks entry, and chronological immutable audit log. | Section Controller on Duty |

## 5. Verification & Physics Compliance Matrix

An automated 10-point test suite (engine/test_suite.py) validates all physical constraints, timetable integrity, and safety laws before any block is granted:

| Test ID | Validation Test Name | Category | Test Verification Result |
|---|---|---|---|
| **1.1** | Geospatial Link Integrity | SQL | **PASSED** (100% assets mapped to valid sections) |
| **1.2** | Timetable Chronology | SQL | **PASSED** (Zero causality violations: exit > entry) |
| **2.1** | Priority Score Bounds | Python | **PASSED** (Scores strictly bounded in [0.0, 100.0]) |
| **2.2** | Execution Completeness | Python | **PASSED** (100% of 220 pending tasks scored) |
| **3.1** | Train Precedence Collision | CP-SAT | **PASSED** (Zero line collisions with moving trains) |
| **3.2** | Resource Capacity Limits | CP-SAT | **PASSED** (Machinery limits strictly respected) |
| **4.1** | Combined Block Structure | CP-SAT | **PASSED** (Multi-dept groups: >=2 depts, aligned) |
| **5.1** | IR Peak Traffic Ban | CP-SAT | **PASSED** (Zero non-critical blocks during 12:00-18:00) |
| **6.1** | Critical Safety Override | Policy | **PASSED** (100% safety tasks locked to Critical >=92.5) |
| **8.1** | Action Workflow Completeness | Actions | **PASSED** (100% tasks mapped to 6-stage action packages) |

## 6. Live System Endpoints & API Access

The system is live, self-contained, and running locally on port 8765:

- **Control Center UI:** http://localhost:8765/
- **System Health Check:** http://localhost:8765/api/health
- **ML Benchmark Leaderboard:** http://localhost:8765/api/ml/benchmark
- **Local SHAP Feature Attribution:** http://localhost:8765/api/ml/explain/<task_id>
- **Deterministic Compatibility Pairs:** http://localhost:8765/api/compatibility
- **Scheduled Blocks & 'Why Chosen':** http://localhost:8765/api/blocks/detailed
- **Government Failure Intelligence:** http://localhost:8765/api/failures
- **Baseline vs CP-SAT Evaluation:** http://localhost:8765/api/evaluation/baseline-vs-optimized
- **Real-Time What-If Simulation:** http://localhost:8765/api/what-if
- **Approval Workflow & Audit Log:** http://localhost:8765/api/approvals & /api/audit